Seismic processing project

B. Arntsen, NTNU

August 26, 2025

Introduction

As part of the course on “Seismic data acquisition and processing” you will process a seismic line, starting with raw data. Also a presentation have to be given documenting the final processing flow and discussing the result. The data is a 2D marine regional line from offshore Japan. The geologic and geophysical setting is documented in the paper by Kodaira et al. (2012, 2017). The acquisition parameters is found in the enclosed paper by Landrø et al. (2019).

You are going to use a commercial processing package, ‘ Reveal” developed by Shearwater Shearwater (2021). The package is installed on a server at IGP, `sand.igp.ntnu.no`. Access to the servers is via ssh login (mac and unix) or via the Xwin32 software running on windows and installed on the machines in the computer Lab in PTS1 at the 4th floor. Login user names and passwords will be distributed when the project work starts. Enclosed with the project is a pdf manual for Reveal. The manual contains a tutorial of the basic functionality for Reveal. There is a data set for the tutorial stored in your home directory.

Scope of project

The project will process a 2D marine dataset from offshore Japan. The processing sequence will include loading of data from a segy file, setting of geometry and processing grid, deconvolution, sorting to cmp gathers, velocity analysis and prestack time migration.

Reveal start up

Reveal is started by typing `reveal` in the terminal window. Your first task is the to create a new project (use the file menu). Assign a descriptive name to the project (f.ex my101) and make sure to thick of the **2D Marine** property. This is important, if not done it will be difficult to create a processing grid.

Input data

The input data is a segy-file found in your home directory /home/¡username¿ with name `my101-2_shot_wotn.sgy` See Hagelund and Levin (2017) for details on the segy format.

Processing sequence

The processing sequence should be as simple as possible and aim at producing pre-stack time migrated data. A flow in Reveal is a sequence of processes, and although it is possible to use one flow for the sequence given below, it is more convenient to use a single flow for each step. Create an initial sequence first to get a first result. Some of the steps given below can be initially skipped and added later.

1. **Input data from the segy file.**
   Use the **Input** tool to read the data and the **Headermath** tool to set source and receiver depths. Then use the **HeaderSetup** tool to compute the `OFFSET, MPT_X, MPT_Y` headerwords. Use the **Output** tool to store the data in the internal seis format. After the data has been loaded you can inspect shots and plot source/receiver coordinates.

2. **Create the processing grid.**
   This step is performed by cliking on the project directory and then edit the processing grid. Select the 'crooked line' geometry and set the cross-line size to at least 960m. Then click on the automatic computation of the processing grid. Check that the processing grid is aligned with the data.

3. **Sort data on shotnumber and receiver number.**
   Use the `Input` tool for sorting on `FFID` and `CHANNEL`. Use the **Header-Setup** tool to compute the `BIN_X, BIN_Y, SRC_CMP, REC_CMP, MPT_ CMP and CMP` headerwords.

4. **Apply Static shift to compensate for source/receiver depths.**
   Use the `HeaderMath` tool to set the `TR_ST` headerwords to the value

   ```
   TR_ST=SRC_ELEV/1500 + REC_ELEV/1500
   ```

   Use the **ApplyStatic** to perform the correction.

5. **Perform Initial velocity analysis.**
   Click on the flow folder and create a velocity analysis flow. You will be guided through a set up sequence. Select to use a picking grid and use every 250 cmp gather for the analysis. After the setup an interactive view with several windows for the velocity analysis will appear. Consult the tutorial for how to pick velocities and store in a table.

6. **Autopick velocities.**
   This step picks velocities for each cmp automatically, but can be skipped in the initial sequence.

7. **Demultiple using radon.**
   This step removes multiples from the data, but can be skipped in the initial sequence.

8. **Deconvolution.**
   This step compresses the pulse and removes the buble, but can be skipped in the initial sequence.

9. **Prestack Time migration.**
   This step performs the imaging of the data. The following tools must be used: (See the Reveal manual and the tutorial) **OffsetbinStack, WindowSort, PrestackTime**.

10. Sort to CMP gathers The output from the prestack time migration is a section for each offset. Sort the output into cmp gathers. The cmp gathers can be used for quality control of the migration.

11. Prestack migration stack.
    use the **Stack** tool to make the final stack.

A tutorial with a similar basic flow is found in the enclosed Reveal manual. The skipped processes can be added afterwards to get an improved result.

Presentation

The results of the project should be given as a power-point presentation. The presentation should cover the following:

- Introduction
- Data
- Methods
- Results
- Discussion and conclusions

and be no longer than 15 minutes. The introduction should contain the geological setting of the area and references to earlier work in the same area. by other authors. Also try to formulate a short objective of the project.

The Data section should contain overview of acquisition parameters, example plots of raw shots, plot of shot positions, and plots of receiver positions for some selected shot points. Try to find shots which illustrates the streamer feathering problem. Results should show plots at various stages of the processing flow and the final section. Discussion and conclusion should mention problems and limitations of the final result.

References

Hagelund, R., and S. A. Levin, 2017, Seg-y revision 2.0 data exchange format 1: Report, Society of Exploration Geophysicists.

Kodaira, S., Y. Nakamura, Y. Yamamoto, K. Obana, G. Fujie, T. No, Y. Kaiho, T. Sato, and S. Miura, 2017, Depth-varying characters in the rupture zone of the the 2011 tohoku-oki earthquake: Geopsphere, **13**, 1408–1424.

Kodaira, S., T. No, Y. Nakamura, T. Fujiwara, Y. Kaiho, S. Miura, N. Takahashi, Y. Kaneda, and A. Taira, 2012, Coseismic fault rupture at the trench axis during the 2011 tohoku-oki earthquake: Nature Geoscience Letters, **5**, 646–650.

Landrø, M., S. Kodaira, T. Fujiwara, T. No, W. Weibull, and B. rge Arntsen, 2019, Time lapse seismic analysis of the tohoku-oki 2011 earthquake: International Journal of Greenhouse Control, **82**, 98–116.

Shearwater, 2021, Shearwater reveal manual: Report, Shearwater Geoservices Software, Inc.